Symmetry Spans and Enforced Gaplessness

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arXiv 2602.11696

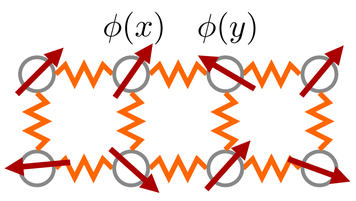
This poster

Overview

Bridging Scales

Constrain IR from UV symmetry. Engineer UV from IR targets.

Microscopic model



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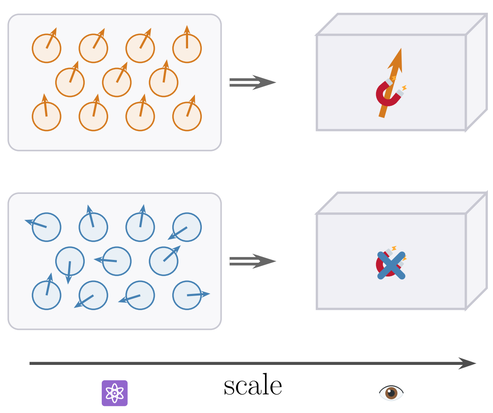
Macroscopic phenomena

gapless (conductor)

gapped (insulator)

Symmetry Classifies Phases

Spins on a lattice with **SO(3) rotation symmetry** can realize different phases:



- **Ferromagnet** : SO(3) spontaneously broken, gapless
- **Paramagnet** : SO(3) preserved, gapped

> ? Can symmetry do more than classify?
> Can it **force** gaplessness — exclude all gapped phases?

Existing Tool: 't Hooft Anomaly Matching

An **anomaly** is a quantum property of a symmetry that persists across scales.

> ! **'t Hooft (1980)**: when a **continuous** symmetry G has an anomaly ($\omega \neq 0$), the IR theory **must be gapless**.

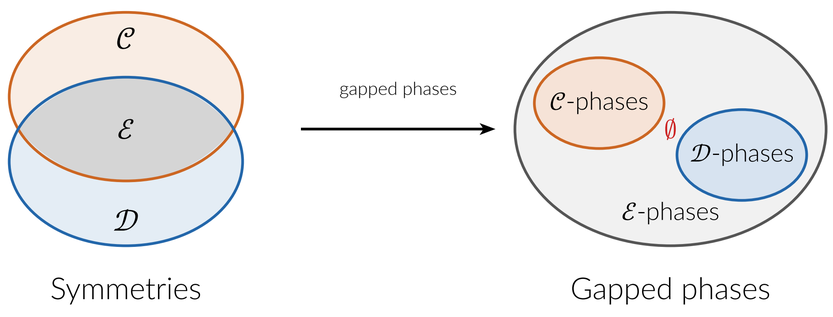
However, perturbative anomalies cannot be realized on a lattice.

> ? Can we find **other** symmetry properties that enforce gaplessness?

New Proposal: Symmetry Span Criterion

A **lattice alternative** to 't Hooft anomaly matching.

Consider **two** symmetries $\mathcal{C}$ and $\mathcal{D}$ acting on a theory, sharing a common sub-symmetry $\mathcal{E}$:



> ! If no gapped $\mathcal{E}$-phase can be simultaneously a $\mathcal{C}$-phase and a $\mathcal{D}$-phase, then the theory must be **gapless**!

In all examples, the span generates a continuous symmetry with anomaly macroscopically.

→ **Lattice alternative of continuous 't Hooft anomaly!**

> ? Systematic constructions and discoveries of gapless lattice systems based on symmetry spans?

Technical Details

The Span Setup: 1+1d, continuous G, finite $H \subset G$

$$\begin{tikzcd} \mathrm{Vect}_H \arrow[r, hook, "i_{\mathcal{C}}"] \arrow[d, hook, "i_{\mathcal{D}}"'] & \mathcal{C} \arrow[d, "g"] \\ \mathbf{Vect}_G^{\omega=0} \arrow[r, "f"'] & \mathcal{S}^{\mathrm{full}} \end{tikzcd}$$

- $\mathcal{D} = \mathbf{Vect}_G^{\omega=0}$: continuous symmetry ($G$ connected)
- $\mathcal{C}$: (possibly non-invertible) finite symmetry
- $\mathcal{E} = \mathrm{Vect}_H$: finite subgroup $H \subset G$
- $\mathcal{S}^{\mathrm{full}}$: Full symmetry generated by $\mathcal{C}$ and G.

Span enforcing gaplessness

- $\mathcal{S}$-symmetric gapped phases $\approx \mathrm{TQFT}(\mathcal{S})$
- $i_{\mathcal{C}}^* : \mathrm{TQFT}(\mathcal{C}) \to \mathrm{TQFT}(\mathcal{E})$: restriction of symmetry

> ! **Criterion:** If $i_{\mathcal{C}}^*\ \mathrm{TQFT}(\mathcal{C}) \cap i_{\mathcal{D}}^*\ \mathrm{TQFT}(G) = \{0\}$, the system admits no symmetric TQFT — it must be gapless.

- $\mathrm{TQFT}(G) \simeq \mathbb{N}[G\text{-SPTs}]$ (no SSB for connected G)

$\mathrm{U}(1) + \mathrm{TY}(\mathbb{Z}_2)$: Kramers–Wannier Duality

- $H = \mathbb{Z}_2, G = \mathrm{U}(1), \mathcal{C} = \mathrm{TY}(\mathbb{Z}_2)$.
- $i_{\mathcal{C}}^*\ \mathrm{TQFT}(\mathrm{TY}(\mathbb{Z}_2)) \ni \left(\mathrm{SSB}_{\mathbb{Z}_2} \oplus \mathrm{triv}_{\mathbb{Z}_2}\right)^n$
  - $\mathrm{SSB}_{\mathbb{Z}_2}$ and $\mathrm{triv}_{\mathbb{Z}_2}$ exchanged by KW duality
- $i_{\mathcal{D}}^*\ \mathrm{TQFT}(\mathrm{U}(1)) \ni \mathrm{triv}_{\mathbb{Z}_2}^m$

> ! Intersection $= \{0\} \Longrightarrow$ **no symmetric TQFT!**

XX Chain (Lattice Realization)

$$\begin{tikzcd} \mathrm{Vect}_{\mathbb{Z}_2} \arrow[r, hook, "i_{\mathcal{C}}"] \arrow[d, hook, "i_{\mathcal{D}}"'] & \mathrm{TY}(\mathbb{Z}_2)_{\mathrm{lattice}} \\ \mathbf{Vect}_{\mathrm{U}(1)}^{\omega=0} & \end{tikzcd}$$

Spin chain $\mathcal{H} = \bigotimes_j \mathbb{C}_j^2$ with:

- KW duality $\mathrm{TY}(\mathbb{Z}_2)_{\mathrm{lattice}}$ and $\mathrm{U}(1)$: $Q = \sum_j X_j$
- $H = \sum_j (Y_j Z_{j+1} - Z_j Y_{j+1}) \overset{\mathrm{RG}}{\rightsquigarrow}$ compact boson at $R = \sqrt{2} R_{\mathrm{s.d.}}$
- Span generates **Onsager algebra** $\overset{\mathrm{RG}}{\rightsquigarrow} \mathrm{U}(1)^{\mathrm{shift}} \times \mathrm{U}(1)^{\mathrm{wind}}$ with mixed anomaly

$\mathrm{U}(1)$-Invariant $\mathbb{Z}_N$ Clock Chain

- $\mathcal{H} = \bigotimes_j \mathbb{C}_j^N$, with $X^N = Z^N = 1$, $XZ = e^{2\pi i/N} ZX$
- Same span structure with $\mathrm{TY}(\mathbb{Z}_N)_{\mathrm{lattice}}$
- Generates **Onsager algebra**
- Symmetric **interacting** Hamiltonian
- RG flow: $c = 1, 3(n-1)/(n+1)$ (integrable)

Outlook

- **Systematic Hamiltonian search:** Find lattice Hamiltonians preserving gaplessness-enforcing spans — *seeking collaborators!*
- **Higher dimensions:** The span criterion extends naturally; e.g. 3+1d Maxwell theory at special coupling.

References